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$$\sum_{n=0}^{\infty} \frac{n^2}{2^n}$$

Klein criterium van Cauchy:

$$\begin{aligned} & \lim_{n \rightarrow +\infty} \sqrt[n]{\frac{n^2}{2^n}} \\ &= \lim_{n \rightarrow +\infty} \frac{\sqrt[n]{n^2}}{2} \\ &= \lim_{n \rightarrow +\infty} \frac{n^{\frac{2}{n}}}{2} \\ &= \frac{1}{2} < 1 \Rightarrow \text{convergent} \end{aligned}$$

References